

1 For the equation to be homogeneous,

$$[v]^2 = m^2 s^{-2} = \left[\frac{a\lambda}{2\pi} \right] = [a][\lambda] \quad \text{and} \quad [v]^2 = m^2 s^{-2} = \left[\frac{2\pi b}{\rho\lambda} \right] = \frac{[b]}{[\rho][\lambda]}$$

Solving the above, we get

$$[a][\lambda] = m^2 s^{-2}$$

$$[a] = m s^{-2}$$

$$\text{and} \quad \frac{[b]}{[\rho][\lambda]} = m^2 s^{-2} \quad [b] = m^2 s^{-2} (\text{kg m}^{-3})(\text{m}) = \text{kg s}^{-2}$$